Tutorial 1- 2026-2027

Exercise 1:

For each of the items listed below, identify which of the 5 components it belongs to and state whether it is classified as a *technological* component or a *less technical/human* component.

Element	Component	Nature
A MySQL database containing customer shipping records.		
A System Analyst designing a new user interface.		
Microsoft Windows 11 running on an office laptop.		
A step-by-step company policy detailing how to process an invoice refund.		
A Solid State Drive (SSD) installed inside a desktop tower.		
Database Administrator (DBA)		
Adobe Photoshop		
Customer shipping address record		
Standard guidelines for handling data breaches		
Network Switch		
Windows Server 2022		
Inventory log spreadsheet file		
IT Helpdesk Technician		

Exercise 2:

Categorize the following software tools into one of the four groups defined in the course text: Operating Systems, Utility Software, Productivity Software, or Programming Software (IDEs).

Software	Category
Asana	
Visual Studio	
Malwarebytes	
Figma	
NetBeans	
WinZip	
Microsoft Teams	
pgAdmin	
ClickUp	

Software	Category
Microsoft Excel:	
Ubuntu Linux:	
Defragment and Optimize Drive	

Exercise 3:

Identify the minimum network communications requirement (No network, Local network, or Internet) for each of the following information system functionalities:

Functionality	Minimum Requirement
Casting a local slide presentation to a smart projector over office Wi-Fi	
Transferring a large dataset via an Ethernet cable to a physical server next to your desk	
Checking real-world currency exchange rates via a live web API widget	
Processing a customer's credit card payment via a cloud gateway terminal (Stripe/PayPal)	
Running a security scan on your computer using locally installed antivirus definitions	
Retrieving shared configuration files from a localized network-attached drive (NAS)	
Hosting a live video webinar with remote clients across the country	
Sending a document to a network-connected office printer down the hallway	
Downloading the latest security patch directly from the Microsoft Update servers	

Exercise 4:

Classify the following devices into the corresponding hardware category (Input, Output, Input-Output, Storage, or Network Communication)

Device	Category
Biometric Fingerprint Scanner	
Network Attached Storage (NAS)	
Interactive Touchscreen Kiosk	
VR Headset (Virtual Reality)	
Laser Receipt Printer	
4G/5G Cellular Modem	

Device	Category
Magnetic Stripe Card Reader	